

Unit 2, Lesson 5: Probability

Benchmarks

MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

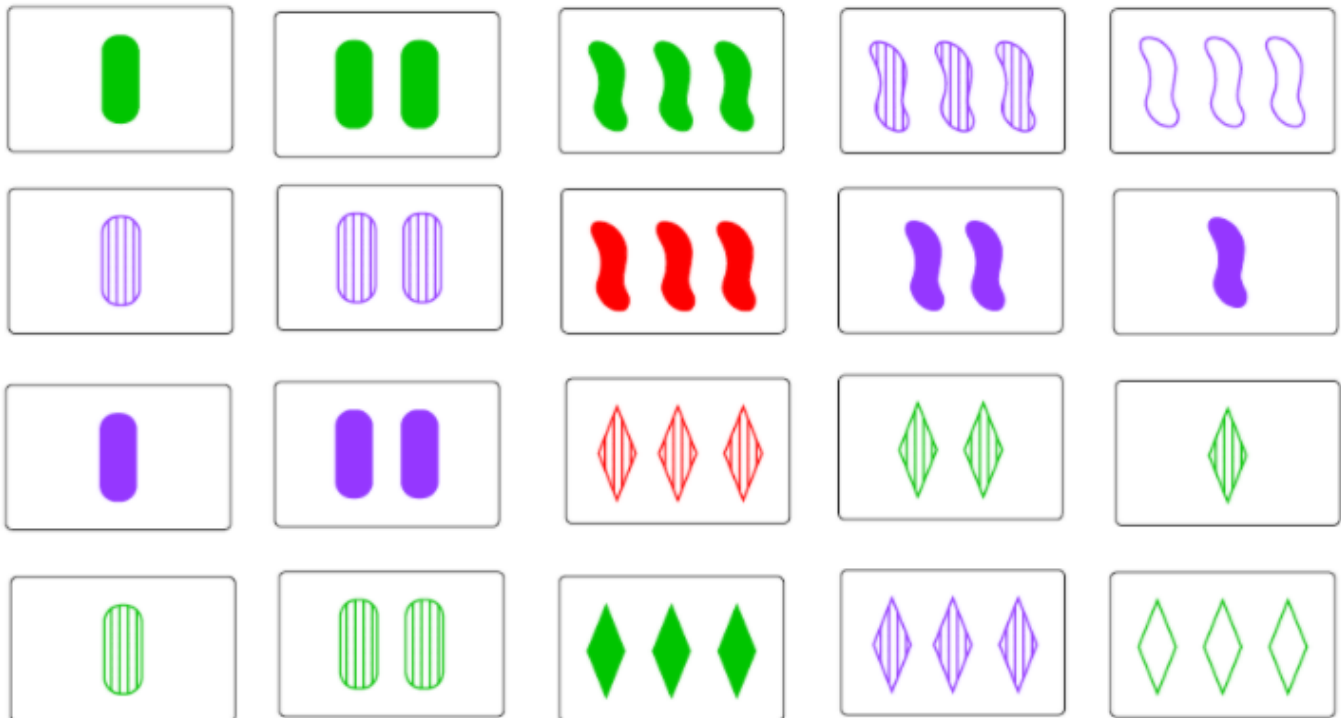
MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

Learning Targets

- ☐ I can express the probability of a chance event using a fraction, percentage, or decimal.
- ☐ I can compare probabilities of chance events.

2.5.1 Warm-Up: Notice and Wonder

1. Some of the cards in the card game SET are shown.



- a. What do you notice?
- b. What do you wonder?

2.5.2 Exploration: Representing Probabilities

When performing simple experiments, all of the outcomes have the same likelihood of happening. For example, when flipping a fair coin, the sample space includes two possible outcomes, {heads, tails}. Each outcome has an equal chance of happening. Therefore, the chance of landing on heads is $\frac{1}{2}$ because this event represents 1 outcome out of 2 possible outcomes. Represented as a decimal or a percentage, the chance of landing on heads is 0.5 or 50%, respectively.

Work with your partner to complete the following.

1. From the percentages listed below, determine the percent chance of each event occurring. Each value is used once.

0%

16. $\bar{6}$ %

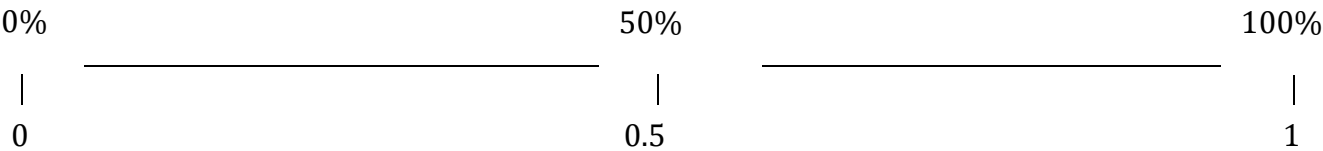
33. $\bar{3}$ %

50%

100%

Event	Percent Chance
a. When given three doors with only one containing a prize behind it, what is the chance of opening the prize-winning door?	
b. When reaching into a dark closet with five pairs of shoes, what is the chance of pulling out a left shoe?	
c. When prompted to randomly select a crayon from a group of red, blue, and yellow crayons, what is the chance you choose a purple crayon?	
d. When rolling a fair six-sided die, what is the chance of landing on 1?	
e. When flipping a coin, what is the chance of landing on either heads or tails?	

2. Plot and label a point to represent the percent chance of each event happening from the previous question. Use the letters labeling each event.





The **probability** of an event is a measure of the likelihood the event will occur. Probabilities are expressed using numbers from 0 to 1 as a **decimal, percentage, or fraction**.

3. Consider the event from question 1 that is least likely to occur.
 - a. Which event is the least likely to happen?
 - b. Describe, using sentences, the likelihood of this event happening.
 - c. Discuss with your partner whether or not it is possible to have a probability less than this.
4. Consider the event from question 1 that is most likely to occur.
 - a. Which event is the most likely to occur?
 - b. Describe, in sentences, the likelihood of this event happening.
 - c. Discuss with your partner whether or not it is possible to have a probability greater than this.
5. Which event has the same likelihood of happening or not happening?

2.5.3 Bring It Together: Representing Probabilities

1. Complete the statements.

The probability of an event can be any number between 0 and 1.

A probability of 0 means it is

- ☐ certain
- ☐ impossible

for the event to occur. A probability of 1 means it is

- ☐ certain
- ☐ impossible

the event will occur. The closer a probability is to 1, the

- ☐ more
- ☐ less

likely it is to

occur, while the closer it is to 0, the

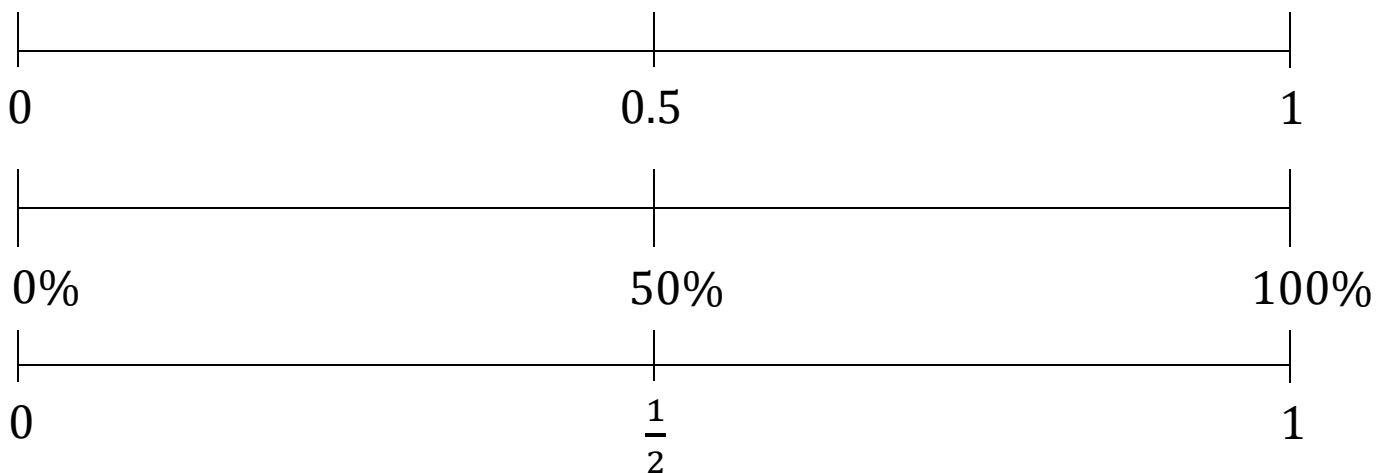
- ☐ more
- ☐ less

likely it is to occur.

A probability of 0.5 means the event is

- ☐ likely.
- ☐ unlikely.
- ☐ neither likely nor unlikely.

Probabilities can be represented by decimals, percentages, or fractions with equivalent values between 0 and 1.



2.5.4 Guided Instruction: Expressing Probabilities of Events



The **probability** of an event occurring, notated as $P(\text{event})$, is determined by dividing the number of outcomes in the event by the possible outcomes in the sample space.

$$P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{total possible outcomes}}$$

Harper is playing a game where she draws a lettered tile out of a bag without looking.

The notation $P(A)$ in this context means “the probability of pulling out a tile with the letter A on it.”

1. Find the probability of each event, represented as a fraction.
 - a. $P(A)$
 - b. $P(B)$
 - c. $P(M)$
 - d. $P(\text{vowel})$
 - e. $P(\text{letter})$



2. Consider the probability of pulling a tile with a consonant on it.
- a. Complete the table to find the fraction, decimal, and percentage equivalents of the probability.

Event	Fraction	Decimal	Percent
$P(\text{consonant})$			

The probability of this event is a repeating decimal. When this occurs, the decimal or percentage representation is typically rounded.

- b. What is $P(\text{consonant})$ represented as a decimal rounded to the nearest hundredth?
- c. What is $P(\text{consonant})$ represented as a percentage rounded to the nearest tenth?

2.5.5 Try It: Expressing Probabilities of Events in Different Forms

1. The probability of rolling a 6 on a standard six-sided die is 16.67%, rounded to the nearest hundredth. Maisie says that this probability is greater than an event whose probability is 0.6 because $16.67 > 0.6$.

Explain whether you agree or disagree with Maisie’s conclusion. Support your argument by showing your work.

2. The probabilities of five events are given.

A	B	C	D	E
40%	0.25	$\frac{5}{6}$	42.9%	$\frac{1}{5}$

- a. Order the values of the probabilities from least to greatest.
- b. Match the probabilities with the events described in the table.

Event		Probability
 Landing on green on the spinner shown		
Pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles		
Landing on a factor of 10 when rolling a fair ten-sided die		
Picking a card with a vowel on it from a set of cards with each letter in the word FLORIDA on it		
Landing on a number less than 6 on a standard six-sided die		

c. Explain which event is the least likely to occur based on the probabilities.

2.5.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can express the probability of a chance event using a fraction, percentage, or decimal.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



2. I can compare probabilities of chance events.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.

